

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1291

South Colton 138 kV
June 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1291. The Interconnection Customer proposes a 91.2 MW large load (data center) facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on September 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 380 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$330,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BOONSBORO to REUSENS 138 kV	BEAR to CLOVER 345 kV L	122.0	388	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BOONSBORO to REUSENS 138 kV	Rebuild line with higher-capacity conductor	\$3,576,000	9% (\$330,000)
	Total allocated to project		\$330,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.